

HR ATTRITION SQL ANALYSIS

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Project Overview

This project analyzes and queries a mock HR dataset using SQL (SQLite) and visualizes data using Power BI. The dataset used in this project contains information about employee attrition, age, income, education, department, several employee satisfaction metrics, and other details.

The objective of this analysis is to identify factors or common trends that may influence employee attrition. Using the available metrics, this project aims to uncover variables and relationships that could be improved to lower employee attrition rate.

Key areas of focus for this analysis are identification of common trends among employees that left, as well as compensation patterns.

Findings

1. Attrition & Employee Profile

- a. Attrition is highest among young, inexperienced employees in entry level roles at 33.12%. Higher attrition is generally linked to lower job levels.
- b. Married employees that have a long commute and travel frequently for business have the highest attrition rate at 21.57% among married employees.
- c. Early tenured employees have the highest attrition rate regardless of career or role stagnation, which may be evidence of early career turnover rather than stagnation.
- d. Among more tenured employees, promotion stagnation is linked to higher attrition rates regardless of role stagnation status.

2. Attrition & Organizational Factors

- a. The Sales department has the highest attrition rate at 20.63%, while the lowest goes to the Research & Development department with an attrition rate of 13.84%.
- b. Attrition is highest among employees with low stock options and low salary hikes at 17.31%. From the rest of the groups, it can be observed that stock option level shows a more consistent link to attrition than salary hike percentage.

3. Compensation Patterns

- a. Using median as a benchmark, there are no notable differences in salary positioning among genders with the exception of Managers and Research Directors. This may be caused by one gender being more skewed towards senior roles compared to the other.
- b. Healthcare Representatives have the widest income range, with most of the other roles closely behind with the exception of Laboratory Technicians and Sales Representatives having a noticeably tighter range.
- c. Monthly income does increase with company tenure for 8 out of 9 roles with Research Directors being the exception, likely due to newer hires holding higher job levels compared

to more experienced employees, though this warrants further investigation due to uneven sample size.

Recommendations

1. Employee Profile & Retention

- a. Strengthening onboarding and early career support for new hires to prevent early career turnover, since attrition is highest among young, inexperienced employees in entry level roles.
- b. Consider more flexible options for business travel or commutes for married employees that have long commutes or frequently travel for business, given their high attrition rate.
- c. Review employee promotion timeline to prevent tenured employees from stagnating too long in one position, since promotion stagnation is linked to notably high attrition among tenured employees.

2. Organizational & Departmental

- a. Investigate potential causes or factors contributing to high attrition in the Sales department when compared to the other departments.
- b. Revisit or prioritize stock option benefits over short term salary hikes when designing contract extensions or offerings, since lower stock option benefits are linked to higher attrition.

3. Compensation Structure

- a. Review compensation practices for Research Director role specifically, because of the atypical manner where newer hires enter at a higher job level than existing tenured employees, though this warrants further confirmation due to uneven sample size.
- b. Investigate drivers behind job level differences by gender for the Research Director and Manager roles, because job level differences are the suspected cause of the salary positioning gap between genders.

Questions and SQL Query List

No	Question	Query	Answer
1	Which department has the highest attrition rate?	<pre>SELECT Department, COUNT(*) AS total_employees, COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS employees_left, ROUND(100.0 * COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) / COUNT(*), 2) AS attrition_rate FROM HR GROUP BY Department</pre>	• The department with the highest attrition rate is the Sales department with 20.63% or 92 out of 446 employees. • The department with the lowest attrition rate is the Research &

		ORDER BY attrition_rate DESC	Development department with 13.84% or 133 out of 961 employees.
2	Are employees that are still young and inexperienced more likely to leave compared to older and more tenured employees?	SELECT CASE WHEN Age < 35 THEN 'Young' ELSE 'Old' END AS age_group, CASE WHEN TotalWorkingYears < 10 THEN 'Inexperienced' ELSE 'Experienced' END AS experience_group, JobLevel, ROUND(100.0 * COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) / COUNT(*), 2) AS attrition_rate, COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count, COUNT(*) AS employee_count FROM HR GROUP BY age_group, experience_group, JobLevel ORDER BY attrition_rate DESC	Yes, the group of employees that are below the age of 35, have less than 10 working years and work in entry roles have the highest attrition rate at 33.12%.
3	Among the married employees that left, is there a pattern of higher time away from home/family compared to married employees that stayed?	SELECT CASE WHEN DistanceFromHome >= 9 THEN 'Far' ELSE 'Close' END AS commute_distance, BusinessTravel, ROUND(100.0 * COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) / COUNT(*), 2) AS attrition_rate, COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count, COUNT(*) AS employee_count FROM HR WHERE MaritalStatus = 'Married' GROUP BY commute_distance, BusinessTravel ORDER BY attrition_rate DESC	- Yes, married employees who have a far commute distance (≥ 9 miles) and frequently go on business trips make up the group with the highest attrition rate at 21.57%. - The highest attrition count belongs to the group that has a far commute but rarely travels and also the group that has a close commute but rarely

			travels. This can be attributed to their higher sample size.
4	Among the employees that left, is there a common pattern of career stagnation?	<pre> SELECT CASE WHEN YearsInCurrentRole > 5 THEN 'Stagnant' ELSE 'Active' END AS role_stagnation, CASE WHEN YearsSinceLastPromotion > 2 THEN 'Stagnant' ELSE 'Active' END AS promo_stagnation, CASE WHEN YearsAtCompany > 5 THEN 'Tenured' ELSE 'New' END AS tenure, ROUND(100.0 * COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) / COUNT(*), 2) AS attrition_rate, COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count, COUNT(*) AS employee_count FROM HR GROUP BY role_stagnation, promo_stagnation, tenure ORDER BY attrition_rate DESC </pre>	- The highest attrition rate (21.09%) occurs among employees that are neither stagnant role nor promotion-wise, however this group is made up of employees that are early within their company tenure. Comparing with the same stagnation profile but with more tenured employees (11.2%), it appears that early career turnover is a more likely driver of attrition. - Among experienced, role-stagnant employees, those who are also promotion stagnant show a higher attrition rate (13.16%) than those who were recently promoted (5.96%), suggesting

			promotion stagnation compounds the risk of role stagnation
5	Are there notable differences in salary positioning between male and female employees performing the same job role?	WITH ranked AS (SELECT JobRole, MonthlyIncome, ROW_NUMBER() OVER (PARTITION BY JobRole ORDER BY MonthlyIncome) AS row_num, COUNT(*) OVER (PARTITION BY JobRole) AS role_total FROM HR), medians AS (SELECT JobRole, AVG(MonthlyIncome) AS median_income FROM ranked WHERE row_num IN ((role_total + 1) / 2, (role_total + 2) / 2) GROUP BY JobRole), employee_salary_position AS (SELECT h.EmployeeNumber, h.JobRole, h.Gender, h.MonthlyIncome, m.median_income, CASE WHEN h.MonthlyIncome < m.median_income THEN 'Below Median' ELSE 'At or Above Median' END AS income_position FROM HR h JOIN	- Using median income as a benchmark, there is no notable difference between genders performing the same job role (within 8% of each other), with the exception of managers (13.81 % difference) and research directors (28.37% difference). - Performing a follow-up job level check shows that a potential reason for the sizable gap is that one gender is more skewed towards senior roles compared to the other.

		<pre> medians m ON h.JobRole = m.JobRole) SELECT JobRole, Gender, median_income, SUM(CASE WHEN income_position = 'Below Median' THEN 1 ELSE 0 END) AS below_count, COUNT(*) AS total_employees, ROUND(SUM(CASE WHEN income_position = 'Below Median' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS below_percentage FROM employee_salary_position GROUP BY JobRole, Gender ORDER BY JobRole </pre>	
6	Is there a common pattern among employees who left, in terms of benefits?	<pre> SELECT CASE WHEN StockOptionLevel < 2 THEN 'Lower' ELSE 'Higher' END AS stock_group, CASE WHEN PercentSalaryHike < 16 THEN 'Lower' ELSE 'Higher' END AS salaryhike_group, ROUND(100.0 * COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) / COUNT(*), 2) AS attrition_rate, COUNT(CASE WHEN Attrition = 'Yes' THEN 1 END) AS attrition_count, COUNT(*) AS employee_count FROM HR GROUP BY stock_group, salaryhike_group ORDER BY attrition_rate DESC </pre>	• The group that consists of employees with lower stock option levels (<2) and lower salary hike percentage (<16%) yielded the highest attrition rate (17.31%) and count (131 employees). • Holding salary hike constant, moving from lower to higher stock options reduces attrition by 5.6-6.9%. Holding stock options constant, moving to a

			higher salary hike only reduces attrition by 0.5-1.9%, showing stock options have a stronger, more consistent link to attrition than salary hike.
7	Which job role has the widest income range?	<pre> WITH mins_and_max AS(SELECT JobRole, MIN(MonthlyIncome) AS min_rate, MAX(MonthlyIncome) AS max_rate FROM HR GROUP BY JobRole) SELECT JobRole, min_rate, max_rate, max_rate - min_rate AS income_range FROM mins_and_max ORDER BY income_range DESC </pre>	- The job role with the widest income range is Healthcare Representatives, with Manufacturing Directors, and Sales Executives close behind with less than \$100 difference. - The rest of the job roles are mostly closely banded together with the exception of Laboratory Technicians and Sales Representatives, which are drastically below the other roles.
8	Does monthly income increase with years at the company? How does that compare	<pre> WITH metrics AS(SELECT JobRole, ROUND(AVG(CASE WHEN YearsAtCompany <= 5 THEN MonthlyIncome END), 2) AS new_income, </pre>	Across 8 out of 9 roles, monthly income does increase with company tenure, with an increase from new employees to

	across job roles?	<pre> ROUND(AVG(CASE WHEN YearsAtCompany > 5 THEN MonthlyIncome END), 2) AS tenured_income, COUNT(*) AS employee_count FROM HR GROUP BY JobRole) SELECT JobRole, new_income, tenured_income, ROUND(tenured_income - new_income, 2) AS difference, employee_count FROM metrics ORDER BY difference DESC </pre>	experienced employees ranging from \$480 to \$1476. • The sole exception is the Research Director role, which has a negative gap of \$928. Performing a follow-up job level check shows that newer hires actually have higher average job levels compared to more tenured employees.
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